1. Write (k_1, v_1)

$$S_1 = \begin{bmatrix}1\\0\end{bmatrix} \begin{bmatrix}3 & 1\end{bmatrix} = \begin{bmatrix}3 & 1\\0 & 0\end{bmatrix}$$

a write is the outer product $k_t v_t^\top$

$$k_1^\top S_1 = \begin{bmatrix}3 & 1\end{bmatrix} = v_1^\top \quad \checkmark$$

the state holds $2 \times 2 = 4$ numbers after every write

2. Write (k_2, v_2)

$$S_2 = S_1 + \begin{bmatrix}0\\1\end{bmatrix} \begin{bmatrix}-1 & 2\end{bmatrix} = \begin{bmatrix}3 & 1\\-1 & 2\end{bmatrix}$$

$$k_1^\top S_2 = \begin{bmatrix}3 & 1\end{bmatrix} = v_1^\top \quad \checkmark$$

$$k_2^\top S_2 = \begin{bmatrix}-1 & 2\end{bmatrix} = v_2^\top \quad \checkmark$$

$k_1 \perp k_2$: each read returns its value exactly

3. Write (k_3, v_3)

$$S_3 = S_2 + \begin{bmatrix}1\\1\end{bmatrix} \begin{bmatrix}2 & 0\end{bmatrix} = \begin{bmatrix}5 & 1\\1 & 2\end{bmatrix}$$

$$k_1^\top S_3 = \begin{bmatrix}5 & 1\end{bmatrix} = v_1^\top + \begin{bmatrix}2 & 0\end{bmatrix}$$

$$k_2^\top S_3 = \begin{bmatrix}1 & 2\end{bmatrix} = v_2^\top + \begin{bmatrix}2 & 0\end{bmatrix}$$

$$k_3^\top S_3 / \|k_3\|^2 = \begin{bmatrix}3 & 1.5\end{bmatrix} = v_3^\top + \begin{bmatrix}1 & 1.5\end{bmatrix}$$

this third key is orthogonal to neither: every read now carries crosstalk

$$k_i^\top S_t = \underbrace{\|k_i\|^2 v_i^\top}_{\text{signal}} + \underbrace{\textstyle\sum_{j \le t,\, j \ne i} (k_i^\top k_j)\, v_j^\top}_{\text{crosstalk}}$$

Every read is free of crosstalk, for any values, only if the keys are orthogonal, and $\mathbb{R}^r$ holds at most r of them (Schlag, Irie and Schmidhuber, 2021). Here $r = d = 2$ and $\phi(k) = k$. In place of the normalizer z_t, a read divides by $\|k_i\|^2$, which is 1 for k_1 and k_2.